

Hierarchical Bayesian Modeling for Improved High-Resolution Mapping of the Completeness Magnitude of Earthquake Catalogs

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Abstract

Assessing the completeness magnitude M_c is essential for most seismicity studies. However, when studying the spatial variation of M_c in a region, the conventional methods that compute M_c based on the frequency–magnitude distribution (FMD) tend to give gaps and large uncertainties of M_c in subregions of low seismicity, thus rendering high-resolution M_c mapping infeasible. To address the limitations of the FMD-based methods, the Bayesian magnitude of completeness (BMC) method was proposed a decade ago to incorporate a priori information about M_c derived from its empirical relationship to the seismic network spatial configuration $M_c = f(d)$, with d being the distance to the k th (typically $k = 4$ or 5) nearest seismic station at each node in space. Although widely used, the BMC method has several critical shortcomings that have long been neglected. In this study, we propose a hierarchical Bayesian model that inherently overcomes these shortcomings of the BMC method for high-resolution M_c mapping coined hierarchical Bayesian magnitude of completeness (H-BMC), which provides a unified and more appropriate approach to the integration of a priori information and local observations concerning M_c . We use an earthquake catalog from the Taiwan region to demonstrate that, compared with the FMD-based methods based solely on observed magnitudes, the proposed H-BMC method effectively utilizes a priori information via prior distributions and thereby gives complete and more reliable high-resolution M_c mapping in terms of gap filling and uncertainty reduction. We also highlight that the H-BMC method for M_c mapping serves as a generic and flexible modeling framework for logically combining imprecise information about M_c from different sources.

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Introduction

The completeness magnitude M_c of an earthquake catalog represents the minimum magnitude threshold above which all earthquake data are considered to be reliably detected and hence describable by the Gutenberg–Richter law (Woessner and Wiemer, 2005; Mignan and Woessner, 2012). Knowledge of the completeness magnitude M_c is essential for virtually any seismicity analysis, such as earthquake forecast, seismic hazard assessment, analysis of rate changes, and static and dynamic triggering. Reliable estimation of M_c is paramount for avoiding bias in the b -value of the Gutenberg–Richter law (Gutenberg and Richter, 1944; Nava *et al.*, 2017), which is often used to extrapolate the probability of potentially damaging earthquakes (Cornell, 1968; Pagani *et al.*, 2020), identify seismic precursors to large earthquakes (Mignan, 2011; Gulia and Wiemer, 2019), and detect stress anomalies and fault asperities (Schorlemmer *et al.*, 2005; Tormann *et al.*, 2014).

M_c is routinely estimated by various methods based on the frequency–magnitude distribution (FMD) of earthquake catalogs, such as maximum curvature (MAXC; Wiemer and Wyss, 2000, actually the mode of the distribution), goodness-of-fit test (Wiemer and Wyss, 2000), M_c by b -value stability (Cao and Gao, 2002), M_c from the entire magnitude range (Woessner and Wiemer, 2005), median-based analysis of the segment slope (Amorè, 2007), and Kolmogorov–Smirnov statistic (Clauzet *et al.*, 2009). However, these conventional methods rely solely

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on FMDs of observed earthquake magnitudes and are subject to several limitations when studying the spatial variation of M_c within a region. One critical limitation is that, in subregions of low seismicity, M_c either cannot be computed due to the requirement for the minimum number of events $N_{\min}$ to construct meaningful local FMDs or may be estimated with great uncertainty because of the small numbers of events; this therefore causes gaps and large uncertainties in spatial estimation of M_c , thus rendering high-resolution mapping of M_c infeasible with the FMD-based mapping methods (Mignan *et al.*, 2011).

To overcome the limitations, Mignan *et al.* (2011) proposed the Bayesian magnitude of completeness (BMC) method that seeks to optimally combine imprecise M_c estimates obtained from observed magnitudes using the FMD-based methods (denoted as M_c^{obs}) with a prior model of M_c based on its empirical relationship to the density of seismic stations. Hence, the BMC method gives improved posterior M_c estimates with reduced uncertainties compared with observed M_c^{obs} , along with gaps being filled by predictions of the M_c prior model. The implementation of the BMC method is available as an R package called rseismNet on GitHub (Mignan, 2019). Over the last decade, the BMC method has been successfully applied in different regions around the globe, such as Taiwan (Mignan *et al.*, 2011), Mainland China (Mignan *et al.*, 2013), Switzerland (Kraft *et al.*, 2013), Lesser Antilles arc (Vorobieva *et al.*, 2013), California (Tormann *et al.*, 2014), Greece (Mignan and Chouliaras, 2014), Iceland (Panzera *et al.*, 2017), South Africa (Brandt, 2019), Texas (Savvaidis *et al.*, 2019), and Venezuela (Vásquez and Guennid, 2013). We are also aware of other ongoing BMC applications in Iran and China regarding a major seismic network update.

For its 10 yr anniversary, we revisit the BMC mapping approach by discussing its critical shortcomings and propose a hierarchical Bayesian model (HBM) that inherently overcomes those for high-resolution M_c mapping, termed hierarchical Bayesian magnitude of completeness (H-BMC). The main advantage of HBM is that it provides a unified and flexible framework for logically combining information from different sources with their uncertainties taken into account (Gelman and Hill, 2006; Carlin and Louis, 2009; Lunn *et al.*, 2012; Gelman *et al.*, 2013; Banerjee *et al.*, 2015; Congdon, 2020). The use of HBM remains relatively recent in seismicity analysis (e.g., Mustać and Tkalčić, 2016; Broccardo *et al.*, 2017; Yang *et al.*, 2019; Himoto, 2020; Shang and Tkalčić, 2020; Rahpeyma *et al.*, 2021) and was not common practice at the time of the development of the original BMC method a decade ago. It therefore appears timely to propose the H-BMC method to address the shortcomings of the original BMC method for improved high-resolution mapping of M_c . To facilitate introduction of the H-BMC method, in the following section we provide a description of the earthquake catalog data used in this article and a brief review of the original BMC method with a discussion on its existing limitations.

Data and the BMC Method

Earthquake catalog

Following the original BMC method (Mignan *et al.*, 2011), in this study, we select the same region of Taiwan, which is known for its high seismicity. We use the earthquake catalog from the Central Weather Bureau Seismic Network (CWBSN) of Taiwan from 1 January 1994 to 31 December 2005, which has been relocated and published in Wu *et al.* (2008). This selected CWBSN earthquake catalog consists of a total of 267,211 seismic events with focal depths as large as around 300 km over a region of approximately 400 km $\times$ 550 km, which were recorded by 76 telemetered stations that are equipped with three-component Teledyne (or Geotech) S13 seismometers. More details regarding this earthquake catalog can be found in the source cited earlier. Figure 1 shows the map and FMD of the entire selected Taiwan CWBSN earthquake catalog.

Review of the BMC method

The basic idea of the BMC method is to use the Bayesian approach to optimally combine a priori information from a generic empirical model of M_c and local M_c estimates (i.e., M_c^{obs}) obtained from observed magnitudes using the conventional FMD-based methods, with both sources of uncertainties taken into account. For a given earthquake catalog of a region, the BMC method generally proceeds in three steps to map M_c in space on a grid.

In the first step, for each grid cell having at least $N_{\min}$ events, a FMD-based method is used to compute M_c^{obs} from the FMD of the observed magnitudes, with their corresponding standard errors σ (i.e., uncertainties) evaluated using the bootstrap method (Efron, 1979). In the original BMC method (Mignan *et al.*, 2011), the mode method (i.e., the so-called MAXC method) was chosen for its robustness in the case of local FMDs (Mignan, 2012; Mignan and Chen, 2016) and for its simplicity in that it only involves finding the magnitude bin with the highest event frequency in the FMD (Wiemer and Wyss, 2000), but it is worth noting that other FMD-based methods can also be used within the BMC framework. In the FMD-based mapping methods, the minimum number of events $N_{\min}$ serves as a threshold to avoid obtaining local M_c^{obs} with unacceptably high uncertainty, and it is thus usually given based on educated guesses, a conservative number ranging from 60 to 500 in the literature (e.g., Wyss *et al.*, 1999; Wiemer and Wyss, 2000; Woessner and Wiemer, 2005; Hutton *et al.*, 2010; Nanjo *et al.*, 2010). As an example, Figure 2 shows the high-resolution M_c map and its associated standard errors σ for the Taiwan CWBSN catalog for the period 1994–2005 computed on a high-resolution $0.1^\circ \times 0.1^\circ$ grid using the mode method and $N_{\min} = 60$. It clearly shows that, although a relatively large $N_{\min}$ can generally guarantee small uncertainties (σ mostly below 0.2) for the computed M_c^{obs} -values, it inevitably results in many gaps (i.e., white area) on the M_c map. On the other

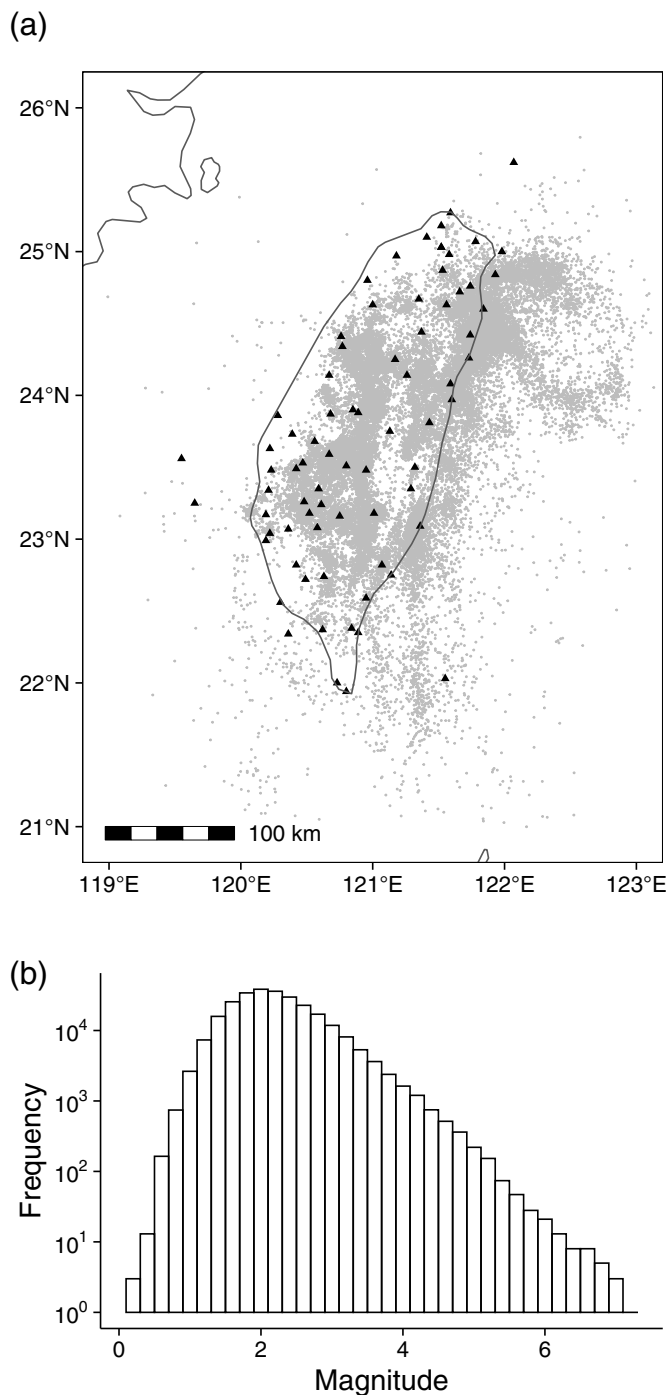


Figure 1. The Taiwan Central Weather Bureau Seismic Network (CWBSN) earthquake catalog from 1 January 1994 to December 31 2005. (a) Map of a random selection of 40,000 seismic events for easier visualization with seismic stations represented by triangle markers. (b) Frequency–magnitude distribution (FMD) for the entire earthquake catalog.

hand, a smaller $N_{\min}$ can certainly reduce gaps on the M_c map, but unfortunately it cannot completely eliminate gaps and leads to large uncertainties of local M_c^{obs} -values for more cells. It is the gaps and large uncertainties associated with the FMD-based

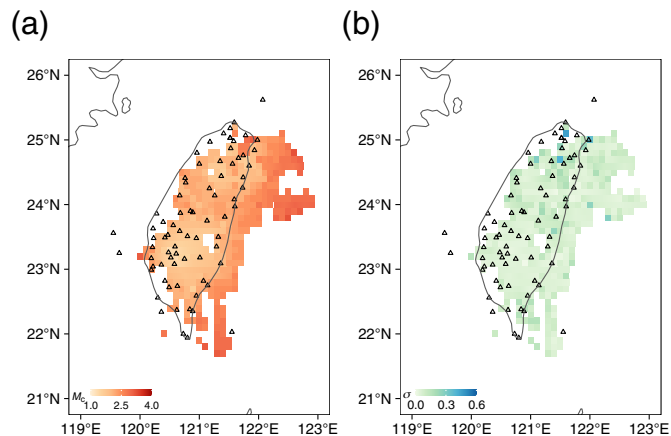


Figure 2. (a) High-resolution M_c^{obs} map and (b) the corresponding standard error σ_{obs} map for the Taiwan CWBSN earthquake catalog of the period 1994–2005 computed on a $0.1^\circ \times 0.1^\circ$ grid using the mode mapping method and $N_{\min} = 60$. The color version of this figure is available only in the electronic edition.

methods in high-resolution M_c mapping that the original BMC method and the H-BMC method proposed in this study seek to address.

For notational convenience, hereafter, we index the observed cells having M_c^{obs} of $i \in 1 : I$ and gap (i.e., unobserved) cells of $j \in 1 : J$. In the second step of the BMC method, a simple model, $M_c = f(d)$, is introduced based on the empirical relationship between M_c and the distance to seismic stations in a network, in which d is the distance to the k th nearest seismic station and k is typically take to be equal to 4 or 5. It is considered that this empirical model can provide a priori information to constrain the uncertain completeness estimates $M_{c(i)}^{\text{obs}}$ from the FMD-based methods. In the original BMC method, the empirical function $f(d)$ is chosen to be the form of $c_1 d^{c_2} + c_3$ with d being the distance to the fifth nearest station. The empirical model is fitted in a nonlinear regression manner to the observed data pairs $(d_i, M_{c(i)}^{\text{obs}})$ of all observed cells i to obtain point estimates of the empirical model parameters, including the standard deviation τ of the fitting residuals under a Gaussian assumption. Then the best-fitting model is used to give a predicted estimate of $M_{c(i)}$ given d_i for each observed cell i , that is, $M_{c(i)}^{\text{pred}} = \hat{c}_1 d_i^{\hat{c}_2} + \hat{c}_3$ with τ considered to be the uncertainty of these predictions.

In the third step of the BMC method, for each observed cell $i \in 1 : I$, the observed and predicted completeness magnitudes (i.e., $M_{c(i)}^{\text{obs}}$ and $M_{c(i)}^{\text{pred}}$) are treated in forms of data likelihood and prior distribution in a Bayesian framework, and by assuming Gaussian likelihood and prior, they are then analytically combined via weighting by their uncertainties to give a posterior mean estimate of the completeness magnitude $M_{c(i)}^{\text{post}}$ as

$$M_{c(i)}^{\text{post}} = \frac{M_{c(i)}^{\text{pred}} \sigma_{\text{obs}(i)}^2 + M_{c(i)}^{\text{obs}} \tau^2}{\tau^2 + \sigma_{\text{obs}(i)}^2}, \quad (1)$$

with the standard deviation $\sigma_{\text{post}(i)}$ of the posterior estimate given by

$$\sigma_{\text{post}(i)} = \sqrt{\frac{\tau^2 \sigma_{\text{obs}(i)}^2}{\tau^2 + \sigma_{\text{obs}(i)}^2}}. \quad (2)$$

For gap cells $j \in 1 : J$ with no observed completeness magnitudes, the posterior estimates and associated standard deviations are simply set to the predicted values from the fitted empirical prior model as

$$\begin{aligned} M_{c(j)}^{\text{post}} &= \hat{c}_1 d_j^{\hat{c}_2} + \hat{c}_3, \\ \sigma_{\text{post}(j)} &= \tau. \end{aligned} \quad (3)$$

Equations (1) and (2) show how uncertain observations and prior predictions of M_c can be combined to generate posterior estimates of M_c with reduced uncertainty, and equation (3) shows how gaps on the M_c map can be filled by predictions of the empirical prior model. Thus, the BMC method achieves complete and more reliable high-resolution mapping of M_c as opposed to the conventional FMD-based methods. However, although simple and useful, the BMC method is subject to several critical shortcomings that have long been neglected as follows:

- When using equation (1) to combine predictions from the empirical prior model with observed $M_{c(i)}^{\text{obs}}$ from local FMDs, the observations $M_{c(i)}^{\text{obs}}$ are actually used twice because the prior model is also calibrated on $M_{c(i)}^{\text{obs}}$. This double use of observations can give biased estimates of the underlying true $M_{c(i)}$ for each cell i .
- The prior model uses point estimates of the empirical parameters (i.e., c_1 , c_2 , and c_3) to generate a priori predictions for M_c and hence fails to account for parameter uncertainty. The disregard for parameter uncertainty will lead to underestimation of the uncertainties in the predictions $M_{c(i)}^{\text{pred}}$ and thus overstatement of the contribution of the prior model in the posterior estimation of M_c .
- For analytical tractability of the posterior estimates $M_{c(i)}^{\text{post}}$, the BMC method is restricted to the Gaussian distributional assumption for both the likelihood and prior models.

These shortcomings of the BMC method arise primarily from the fact that the two likelihood and prior models of M_c are handled separately rather than holistically in a fully Bayesian approach. In this study, we propose a unified and flexible HBM method that inherently overcomes the shortcomings of the BMC method for high-resolution mapping of M_c and demonstrate how the proposed H-BMC method can logically combine imprecise information from the observations and empirical model of M_c , thereby achieving improved spatial estimation of M_c compared with the conventional FMD-based methods.

HBM for M_c Mapping Model formulation

The novelty of the H-BMC method over the simpler BMC method is that the true completeness magnitude $M_{c(i)}$ for each cell i is modeled as a latent variable, which is informed simultaneously by the likelihood and prior models. For the purpose of simplicity and demonstration, here we follow the BMC method and continue using Gaussian distribution for the likelihood and prior, but we emphasize that the H-BMC method is not restricted to any particular type of distributional assumption. As denoted previously, we index the observed and gap cells by $i \in 1 : I$ and $j \in 1 : J$, respectively. Suppose that we already have a map of local completeness magnitude estimates $M_{c(i)}^{\text{obs}}$ obtained from the observed magnitudes of an earthquake catalog using a chosen FMD-based method, along with the associated standard errors $\sigma_{\text{obs}(i)}$. Further, we choose to use the empirical relationship $M_c = f(d) = c_1 \log_{10} d + c_2$ instead of $M_c = c_1 d^{c_2} + c_3$ as the prior model because it is simpler but with a similar functional form; moreover, a logarithmic function is consistent with the seismic waveform attenuation functions commonly used in seismic hazard. Although the main purpose of this article is to propose the H-BMC method to serve as a generic modeling framework that logically combines imprecise information about M_c from different sources for improved spatial estimation, we, for illustrative purpose, provide an informal comparison of the two aforementioned empirical functional forms later in the Discussion section and demonstrate the superiority of our employed log-form prior model. The HBM for M_c mapping is formulated as follows.

For each observed cell i , the likelihood function of the unknown true completeness magnitude $M_{c(i)}$ for the observed $M_{c(i)}^{\text{obs}}$ is expressed as

$$\begin{aligned} p(M_{c(i)}^{\text{obs}} | M_{c(i)}, \sigma_{\text{obs}(i)}) &= \text{Normal}(M_{c(i)}, \sigma_{\text{obs}(i)}^2) \\ &= \frac{1}{\sqrt{2\pi}\sigma_{\text{obs}(i)}} \exp\left(-\frac{(M_{c(i)}^{\text{obs}} - M_{c(i)})^2}{2\sigma_{\text{obs}(i)}^2}\right). \end{aligned} \quad (4)$$

In the HBM framework, to achieve statistical shrinkage of the parameters of interest, parameters (i.e., $M_{c(i)}$ in our case) are assigned priors with common hyperparameters (i.e., parameters of priors) based on a flexible assumption termed exchangeability—that is, the joint probability distribution of a sequence of random variables does not depend on the order of the variables. This exchangeability assumption clearly holds for $M_{c(i)}$'s given their empirical relationship to the density of seismic stations $M_c = f(d)$.

Thus, the prior distribution for each $M_{c(i)}$ that embodies the empirical relationship $M_c = f(d)$ with common hyperparameters $\{c_1, c_2, \tau\}$ is given as

$$p(M_{c(i)}|d_i, c_1, c_2, \tau) = \text{Normal}(c_1 \log_{10}(d_i) + c_2, \tau^2) \\ = \frac{1}{\sqrt{2\pi}\tau} \exp\left(-\frac{(M_{c(i)} - c_1 \log_{10}(d_i) - c_2)^2}{2\tau^2}\right). \quad (5)$$

Then, the three unknown hyperparameters $\{c_1, c_2, \tau\}$ are assigned hyperpriors (also called hierarchical priors). Because here we have no prior knowledge about the specific values of these hyperparameters, vague or weakly informative priors that span a relatively wide range of parameter values can be specified, and they generally allow the data to dominate the posterior inference. Here, we give the weakly informative priors for $\{c_1, c_2, \tau\}$ as follows:

$$\begin{aligned} p(c_1) &= \text{Normal}(0, 10) \\ p(c_2) &= \text{Normal}(0, 10) \\ p(\tau) &= \text{Half-normal}(0, 5) \end{aligned} \quad (6)$$

Half-normal distribution can be thought of as a normal distribution truncated below zero. It is noteworthy that alternative formulations of the previous weakly informative priors are possible for the HBM, as is always the case in Bayesian inference. An informal prior sensitivity analysis (not reported here for brevity) suggests no difference in the posterior inference when alternative vague and weakly informative priors are employed.

For notational convenience, we use $\mathbf{y}_i = \{d_i, M_{c(i)}^{\text{obs}}, \sigma_{\text{obs}(i)}\}$ to denote all observed data associated with cell i and $\mathbf{y}_{1:I}$ to denote $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_I\}$. According to Bayes' theorem, the joint posterior distribution of all unknown parameters $\{M_{c(1:I)}, c_1, c_2, \tau\}$ given all data $\mathbf{y}_{1:I}$ is expressed as

$$p(M_{c(1:I)}, c_1, c_2, \tau | \mathbf{y}_{1:I}) \\ = \frac{p(c_1)p(c_2)p(\tau) \prod_{i=1}^I p(M_{c(i)} | d_i, c_1, c_2, \tau) \prod_{i=1}^I p(M_{c(i)}^{\text{obs}} | M_{c(i)}, \sigma_{\text{obs}(i)})}{p(M_{c(1:I)}^{\text{obs}} | d_{1:I}, \sigma_{1:I})} \\ \propto p(c_1)p(c_2)p(\tau) \prod_{i=1}^I p(M_{c(i)} | d_i, c_1, c_2, \tau) \prod_{i=1}^I p(M_{c(i)}^{\text{obs}} | M_{c(i)}, \sigma_{\text{obs}(i)}), \quad (7)$$

from which the marginal posterior distribution of each parameter (e.g., $p(M_{c(i)} | \mathbf{y}_{1:I})$) can be obtained via marginalizing over the posteriors of all the parameters and the posterior mean and credible interval (CI; or standard deviation) may be respectively summarized as the point estimate and associated uncertainty, respectively.

Equations (4)–(7) constitute the HBM for spatial estimation of M_c . After the HBM is fitted, predictions can be made for gap cells j about their true completeness magnitude $M_{c(j)}$ given the respective observed distance to the k th nearest station d_j . The posterior predictive distribution of $M_{c(j)}$ for each new cell j , $p(M_{c(j)} | d_j, \mathbf{y}_{1:I})$, is expressed as

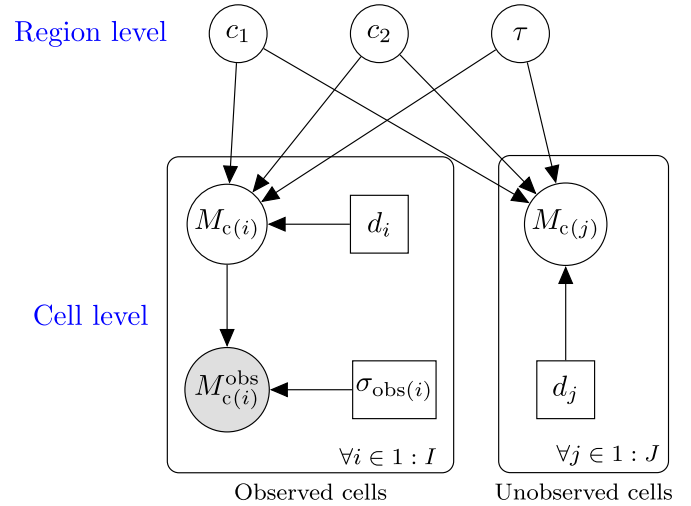


Figure 3. Directed acyclic graph of the proposed hierarchical Bayesian magnitude of completeness (H-BMC) method for M_c mapping. Nodes of open circles, filled circles, and rectangles indicate unobservable variables, observable variables, and constants, respectively; directed edges indicate conditional dependence; and plates indicate repetition for what is inside it. The color version of this figure is available only in the electronic edition.

$$p(M_{c(j)} | d_j, \mathbf{y}_{1:I}) \\ = \int \int \int p(M_{c(j)} | d_j, c_1, c_2, \tau) p(c_1 | \mathbf{y}_{1:I}) p(c_2 | \mathbf{y}_{1:I}) p(\tau | \mathbf{y}_{1:I}) d\Delta_1 d\Delta_2 d\tau, \quad (8)$$

in which $p(M_{c(j)} | d_j, c_1, c_2, \tau) = \text{Normal}(c_1 \log_{10} d_j + c_2, \tau^2)$.

Figure 3 illustrates the graphical representation of the proposed H-BMC method for M_c mapping using the directed acyclic graph in plate notation. It shows that treating the true completeness magnitudes $M_{c(i)}$ as latent variables in hierarchical modeling, the unknown parameters of the empirical prior model at the higher regional level are indirectly informed by all local observations $M_{c(i)}^{\text{obs}}$ at the lower cell level, which in turn constrain all cell-level parameters $M_{c(i)}$ in conjunction with observations $M_{c(i)}^{\text{obs}}$. These structural dependencies across parameters provide better informed estimates for all the parameters. Hence, the posterior distribution $p(M_{c(i)} | \mathbf{y}_{1:I})$ reflects a logical combination of imprecise information about $M_{c(i)}$ from the local observation of $M_{c(i)}^{\text{obs}}$ at the cell level and the empirical prior model at the regional level.

Posterior distributions in Bayesian inference usually do not have an analytical form and are thus commonly numerically approximated using Markov chain Monte Carlo (MCMC) simulation methods, especially for complex hierarchical models. In this study, all Bayesian computations are performed using the modern Bayesian inference program Stan (Stan Development Team, 2021), which implements two MCMC methods called the Hamiltonian Monte Carlo (HMC) algorithm and the

No-U-Turn sampler of the HMC's adaptive variant. A detailed introduction to these MCMC algorithms is beyond the scope of this article but can be readily found elsewhere (e.g., Brooks *et al.*, 2011; Neal, 2011; Hoffman and Gelman, 2014; Betancourt, 2019).

The model formulation presented earlier demonstrates that the proposed H-BMC method inherently overcomes the shortcomings of the original BMC method for spatial M_c mapping discussed previously. First, under the hierarchical framework, the prior model parameters are estimated simultaneously with M_c of all cells (see Fig. 3 and joint posterior of equation 7), thus avoiding double use of observations $M_{c(i)}^{\text{obs}}$ when incorporating a priori information. Second, the uncertainties of the empirical parameters are inherently accounted for in the posterior estimates of M_c ; as in a fully Bayesian approach, the marginal posterior distribution of each parameter is obtained via marginalizing over the posterior distributions of all the other parameters. Finally, because we use numerical methods, MCMC simulation methods, to implement posterior computation, there are essentially no restrictions on the type of distributions for the likelihood and prior models.

It is worth noting that, although both the BMC and its improved H-BMC mapping methods are presented in a 2D fashion for the moment, there is generally no model bias expected due to earthquake depths z in the case in which the seismicity depth profile is consistent throughout the region considered (because $M_c(z)$ variations have then the same impact on the shape of the FMD per (x,y) cell). This is the case in the present study in which only crustal seismicity is considered. Earthquake depths should be taken into account when both crustal and subduction regimes are involved in the study or when variations along the three dimensions are of the same order, such as in a local seismicity cloud in the induced seismicity regime (see the 3D prior model in Mignan, 2021). In other cases, variations of $M_c(x,y)$ predominate over potential $M_c(x,y,z)$ variations.

Results

We choose the mode method as an example benchmark to compute local M_c^{obs} from FMDs of observed magnitudes and demonstrate how the proposed H-BMC method improves spatial mapping of M_c via incorporating a priori information about M_c compared with the conventional FMD-based methods. The minimum number of events N_{min} for M_c computation is set to a small number of 10 rather than a large number of over 60, which allows us to clearly see large uncertainties in local M_c estimates obtained solely from observed magnitudes using FMD-based methods. In addition, we use the distance d to the fourth (i.e., $k = 4$) nearest station for the empirical prior model $M_c = c_1 \log_{10} d + c_2$. The choice of k will be discussed later.

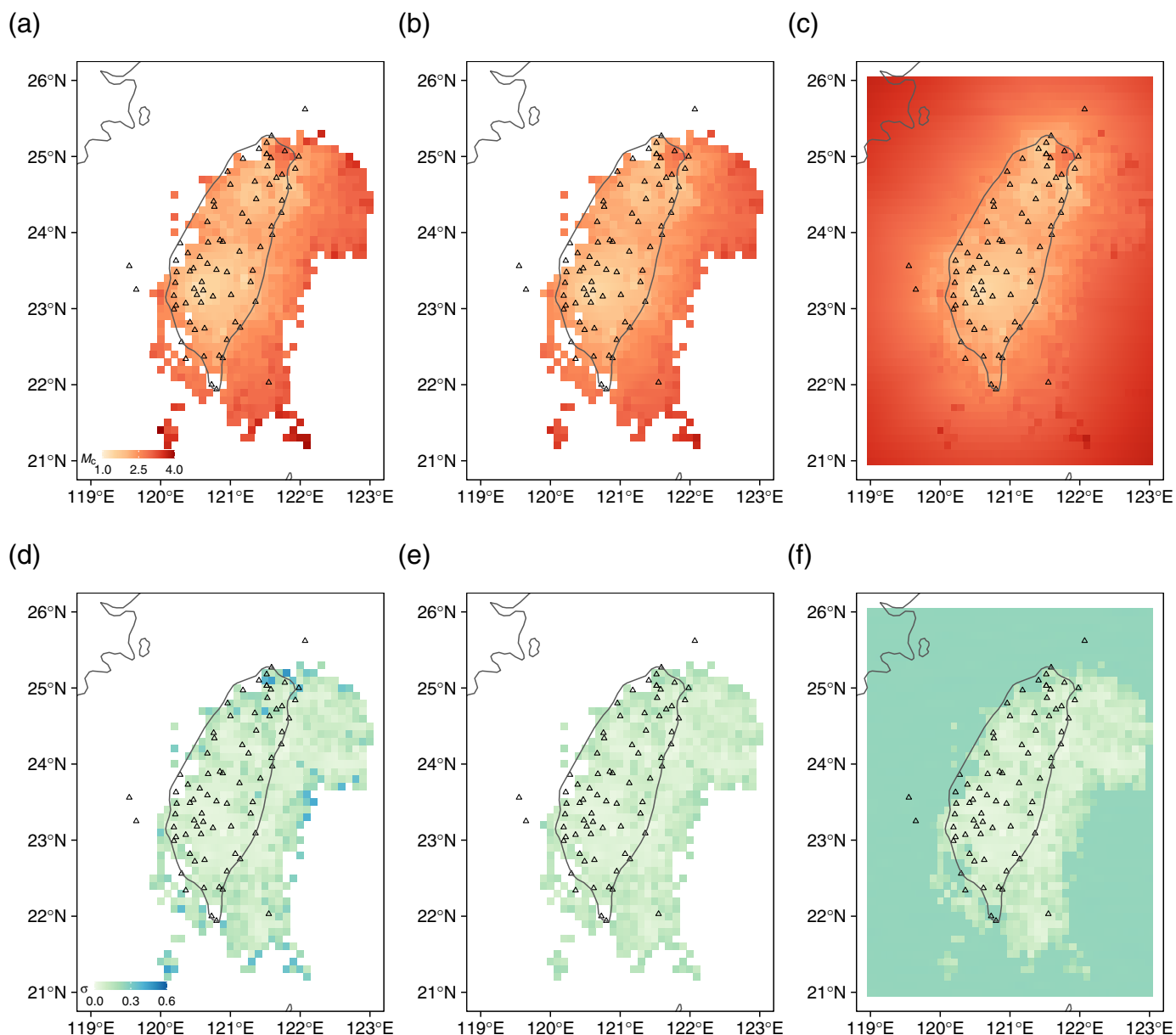
Figure 4a,d depicts the high-resolution maps ($0.1^\circ \times 0.1^\circ$) of the completeness magnitudes M_c and standard errors σ obtained

from observed magnitudes of the Taiwan CWBSN catalog using the mode method with $N_{\text{min}} = 10$. These two maps show that, compared with Figure 2 with $N_{\text{min}} = 60$, whereas the area over which we could obtain M_c estimates is greatly expanded, the uncertainties are exacerbated with many cells having σ as large as nearly 0.6. It is this irreconcilable conflict between mapping integrity and reliability that renders high-resolution M_c mapping using the FMD-based methods practically infeasible.

Now, we demonstrate how the H-BMC method described by equations (4)–(8) can give improved high-resolution mapping of M_c compared with the FMD-based mode method. First, we give posterior inferences of the empirical model parameters (c_1 , c_2 , and τ). Figure 5a shows the trace plots of the three Markov chains (4000 posterior MCMC draws for each chain after the first 2000 draws were discarded in the burn-in period) for the three empirical parameters, and the three well-mixed and stable chains visually indicate convergence of these simulated draws to their target posterior density. On a side note, in our particular analysis case, MCMC sampling converges after around 500 iterations in the burn-in period. Figure 5b shows the histograms of these simulated posterior draws used for approximating the posterior distributions of the parameters, from which any posterior statistics, such as mean, standard deviation, and CIs, can be readily summarized. Note that, in Bayesian inference, the posterior mean is taken to be a point estimate, and posterior standard deviation and CI are taken to be a measure of posterior uncertainty. Table 1 reports the mean, standard deviation, and 95% CI of the posterior distributions of the three empirical parameters. It shows that these parameters are indeed subject to some degree of uncertainty, particularly for c_2 , which however is ignored in the original BMC method and can thus lead to biased local M_c estimates.

The posterior means M_c^{post} and standard deviations σ_{post} of M_c for observed and unobserved cells can be summarized from their respective posterior distributions $p(M_{c(i)}|y_{1:i})$ and $p(M_{c(j)}|d_j, y_{1:i})$, respectively. Figure 4b,e illustrates the posterior M_c and σ maps for observed cells, respectively, and they clearly show an overall substantial reduction of uncertainties in comparison with their counterpart of Figure 4a,d after incorporating information from the empirical prior model of M_c . Notably, all extreme σ -values over 0.3 for M_c^{obs} are practically eliminated in the posterior map of M_c^{post} . Figure 4c,f shows the complete posterior high-resolution maps of M_c and σ with predictive estimates for unobserved cells added. The gaps in M_c mapping are filled by the posterior predictive estimates M_c^{post} resulting from the proposed H-BMC method, which largely fall between 2.0 and 3.5 with the associated posterior standard deviations σ_{post} between 0.24 and 0.25.

A better visual comparison between the observed and posterior M_c and their uncertainties for observed cells is presented in Figures 6 and 7. Figure 6 shows the best-fitting empirical prior model for M_c against the distance d to the fourth nearest

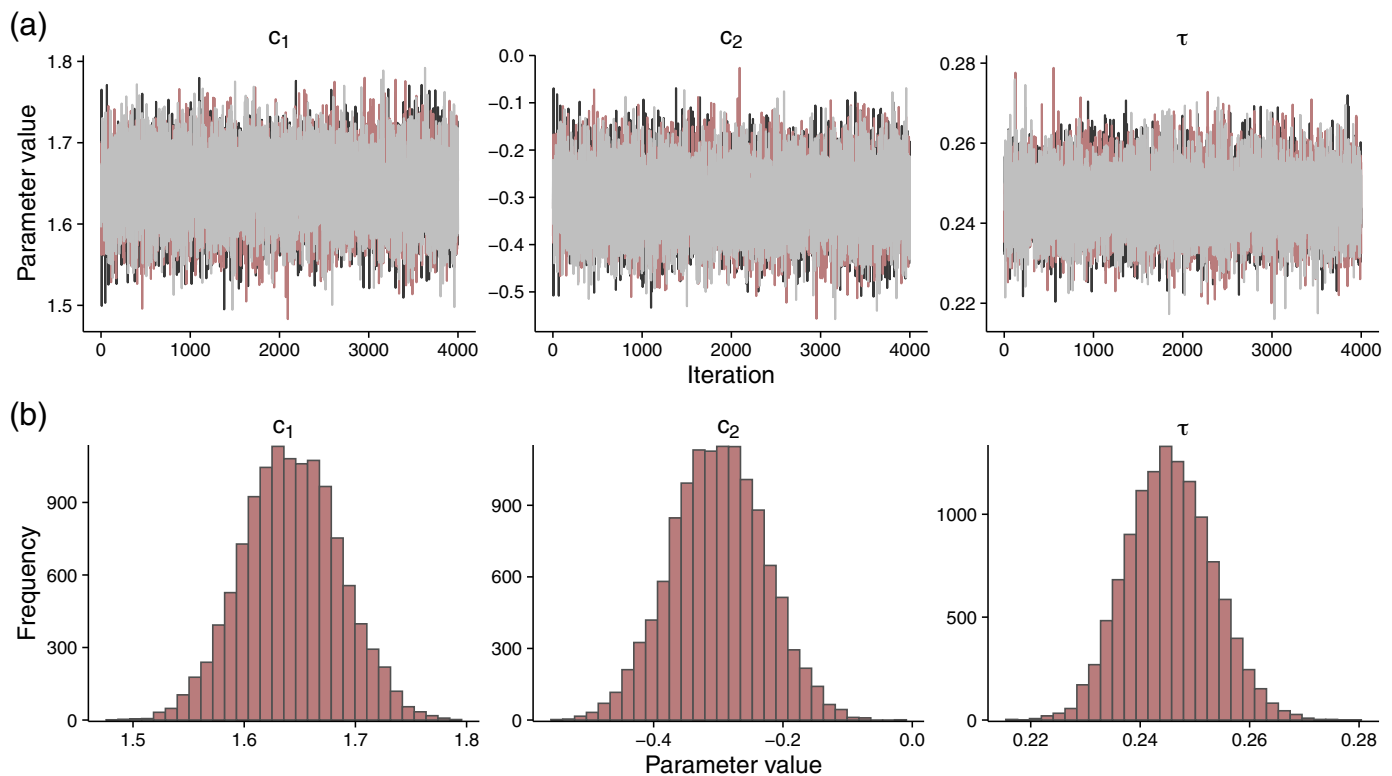


station, together with M_c^{obs} and M_c^{post} overlaid. It can be seen that, when the higher-level prior information is introduced within the hierarchical model, the M_c^{post} estimates for the observed cells are pulled closer together around the best-fitting empirical model than the M_c^{obs} estimates, indicating reduced variation between the lower cell-level parameters. This shrinkage (sometimes called pooling) of posterior estimates at the lower level toward the higher-level mean (i.e., the mean line of the region-level empirical model) accompanied by uncertainty reduction is the general property of hierarchical modeling (Gelman and Hill, 2006; Carlin and Louis, 2009; Gelman *et al.*, 2013; Kruschke, 2014), and it occurs because of the influence from two information sources at the cell level and regional level.

The scatter plot between M_c^{obs} and M_c^{post} in Figure 7a clearly shows that, after applying the H-BMC method to incorporate

Figure 4. Bayesian hierarchical high-resolution mapping of the completeness magnitude of the Taiwan CWBSN catalog for the period 1994–2005 on a $0.1^\circ \times 0.1^\circ$ grid. (a) Local M_c estimates (i.e., M_c^{obs}) obtained from observed magnitudes using the mode method and (d) associated standard errors σ_{obs} . (b) Posterior means of M_c (i.e., M_c^{post}) and (e) associated standard deviations σ_{post} for the observed cells after incorporating information from the empirical prior model of M_c . (c) Posterior (predictive) means M_c^{post} for both the observed cells and unobserved (i.e., gap) cells and (f) associated standard deviations σ_{post} . The color version of this figure is available only in the electronic edition.

a priori information about M_c , the majority of observed estimates M_c^{obs} are corrected (i.e., shrunk) downward to arrive at smaller values of posterior estimates M_c^{post} , as indicated by most data points lying below the blue dotted reference line of equality.



This implies that, for this particular Taiwan CWBSN catalog, the chosen FMD-based mapping method tends to overestimate M_c in space. The scatter plot between σ_{obs} and σ_{post} in Figure 7b shows the aforementioned uncertainty reduction accompanied with shrinkage when hierarchical modeling is applied, and the large deviation of many data points below the reference line of equality indicates that many observed estimates M_c^{obs} were significantly improved.

The results discussed thus far demonstrate that the proposed H-BMC mapping method can give complete and more reliable high-resolution M_c mapping compared with the conventional FMD-based methods that rely solely on observed magnitudes of earthquake catalogs. We also discussed how the H-BMC method inherently overcomes the critical shortcomings of the original BMC method in the formulation of the HBM for spatial M_c mapping.

TABLE 1
Posterior Summary Statistics of the Empirical Model Parameters

Parameter	Mean	Standard Deviation	95% CI
c_1	1.643	0.044	(1.557, 1.728)
c_2	-0.301	0.072	(-0.443, -0.158)
τ	0.246	0.008	(0.231, 0.261)

CI, credible interval

Figure 5. Posterior Markov chain Monte Carlo (MCMC) draws of the three empirical model parameters: (a) trace plots of the three Markov chains in different colors and (b) histograms of the posterior draws. Here, a large number of 12,000 posterior draws in total are simulated via MCMC, which is a typical number for complex Bayesian models. The color version of this figure is available only in the electronic edition.

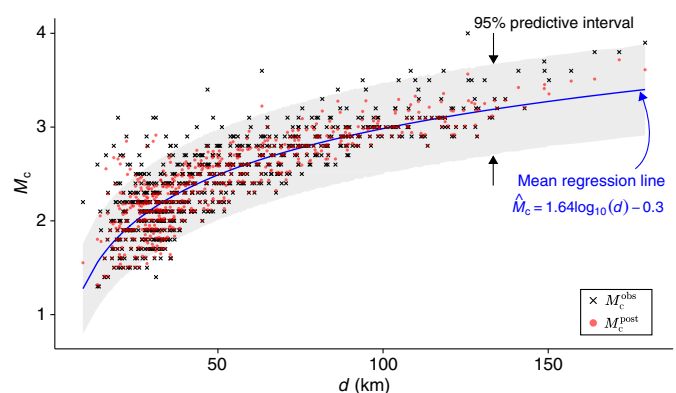


Figure 6. Comparison of observed and posterior completeness magnitudes, M_c^{obs} and M_c^{post} , along with the best-fitting empirical prior model $M_c = 1.64 \log_{10}(d) - 0.3$ within the hierarchical Bayesian model. The 95% predictive interval represents the range within which the predicted M_c will lie for a given d with 0.95 probability and is calculated by taking into account posterior uncertainties of all model parameters. The color version of this figure is available only in the electronic edition.

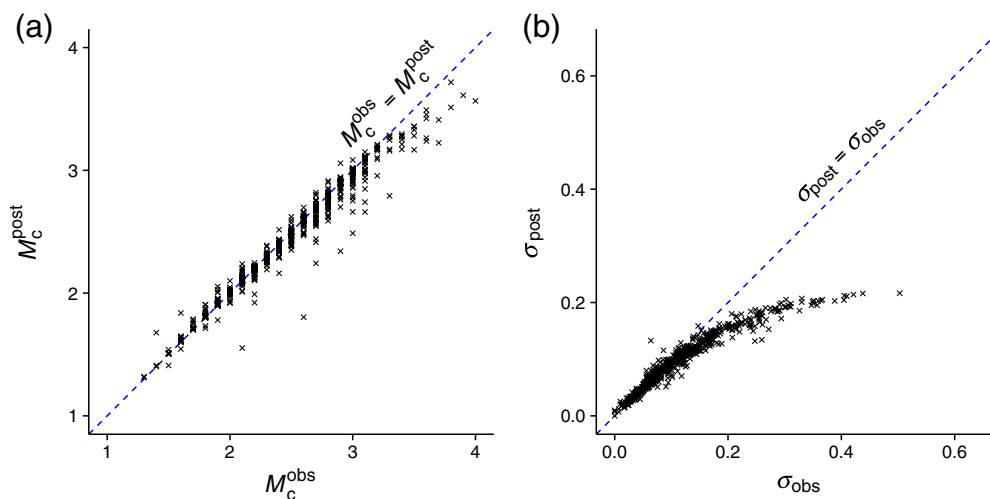


Figure 7. Scatter plot for (a) M_c^{post} against M_c^{obs} and (b) σ_{post} against σ_{obs} . The color version of this figure is available only in the electronic edition.

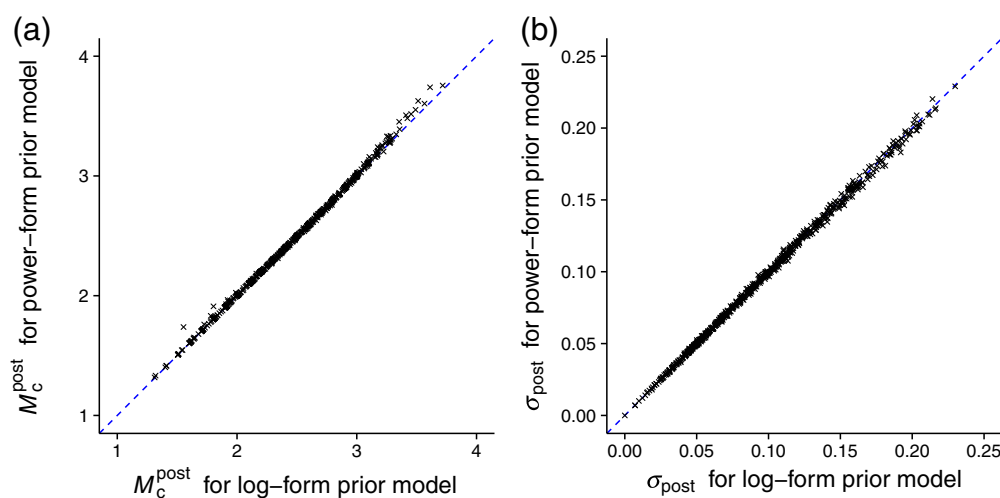


Figure 8. Comparison between the log-form and power-form empirical prior models for (a) posterior M_c estimates M_c^{post} and (b) the associated posterior standard deviation σ_{post} . The color version of this figure is available only in the electronic edition.

Discussion

The main goal of this article is to propose the H-BMC method that overcomes the critical shortcomings of the widely used BMC method in spatial mapping of M_c and serves as a generic modeling framework for logically combining imprecise information about M_c from observed FMDs and other a priori sources (e.g., empirical relationships) with their respective uncertainties taken into account. As the aphorism goes, all models are wrong, but some are useful (Box, 1976). Although we have demonstrated that the proposed H-BMC method can reduce uncertainties of M_c estimates and fill M_c gaps resulting from conventional FMD-based methods (also the purpose of the

original BMC method), the specific choices in several model aspects (e.g., assumptions), similar to any statistical modeling, may be refined within the H-BMC framework to further improve its performance on spatial estimation of M_c . To give some insights, we subsequently discuss a number of model aspects that may deserve future investigation for continuous model improvement within our proposed H-BMC framework.

Functional form of the empirical relationship

Although the prior model of M_c is based on its empirical relationship to the density of seismic stations in a network, currently there is no universally agreed-upon functional form for this empirical relationship. The original BMC method used the power-form $M_c = c_1 d^{c_2} + c_3$, and for the moment we suggest using the simpler log-form $M_c = c_1 \log_{10} d + c_2$ for the reasons discussed in the [Model formulation](#) section. The empirical model of M_c at the regional level may be further refined in the future when more empirical evidence is accumulated from statistical analysis of different earthquake catalogs.

As an illustrative example,

Figure 8 shows an indistinguishable difference in both the posterior estimates M_c^{post} and their uncertainties σ_{post} between the log-form (i.e., $M_c = c_1 \log_{10} d + c_2$) and power-form (i.e., $M_c = c_1 d^{c_2} + c_3$) prior models. This informally suggests that, for the particular CWBSN dataset, the simpler log-form empirical relationship should be favored as a prior model choice considering that more complex models tend to generalize worse than the simpler models (i.e., overfit the data at hand). Nevertheless, the decision of which empirical functional form of M_c to use should be subjected to formal model comparison depending also on the underlying structure of the earthquake catalog used.

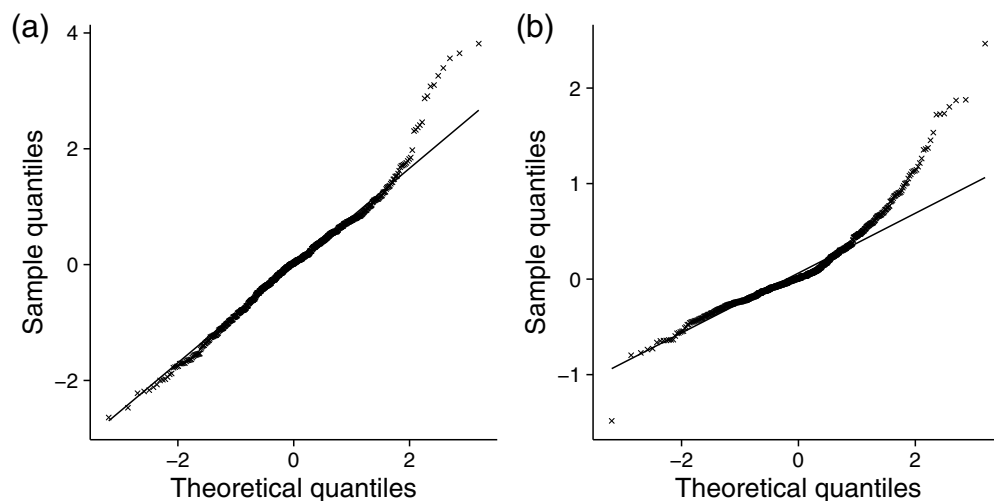


Figure 9. Normal Q-Q plot of posterior mean standardized residuals of M_c for (a) the prior model of equation (5) at the regional level and (b) the likelihood model of equation (4) at the cell level.

Distance d to the k th nearest station

In this study, we used the distance d to the fourth nearest station ($k = 4$) as the default choice for the empirical prior model $M_c = f(d)$. Although Mignan *et al.* (2011) demonstrated the original BMC method using $k = 5$, they noted that using $k = 3, 4$, or 5 may be reasonable, and they verified that, for the particular Taiwan CWBSN dataset, these three k -values showed indistinguishable differences in posterior M_c mapping. Notwithstanding, when analyzing other earthquake catalogs, different k -values may be examined to determine the optimal k that gives best-posterior performance. In addition, although the distance to the k th nearest station is indeed a simple proxy for the local station density, it provides an effective way to encode a priori information about M_c from the seismic network spatial configuration.

Distributional assumption

As with the BMC method and for the ease of demonstration, Gaussian errors are assumed in both the likelihood and prior models for the H-BMC method, yet other distributional assumptions can also be explored. As an example, Figure 9a,b gives the normal Q-Q plot of the posterior mean standardized residuals of M_c for the prior model of equation (5) and likelihood model of equation (4), respectively. The two subfigures show that, at the regional level, the Gaussian error assumption for the prior model is generally adequate except for a few extreme values in the upper tail, but this assumption appears inadequate for the likelihood model at the cell level given a large portion of standardized residuals in the upper tail deviating significantly from the reference line of equality. This is indicative of some room for model improvement by, for example, using a distribution that has a heavier right tail. To further complicate this matter, spatial correlation and error

heteroscedasticity may also be considered in the distributional assumptions.

Bayesian model comparison

The aspects for model criticism discussed earlier all require ongoing formal model comparison and selection. In the fully Bayesian framework, the two advocated formal methods are the widely applicable information criterion and leave-one-out cross validation, which compare the out-of-sample predictive accuracy of models based on parameters' full posterior distributions rather than point estimates only (Vehtari

and Ojanen, 2012; Gelman *et al.*, 2013, 2014; Spiegelhalter *et al.*, 2014; Piironen and Vehtari, 2017; Vehtari *et al.*, 2017). However, investigating the complex problem of Bayesian model comparison and selection is beyond the scope of this article and is worth future efforts.

We emphasize that the proposed H-BMC method is not restricted to any particular type of (1) FMD-based method, (2) empirical functional form relating M_c to the station density d , (3) distance to the k th nearest station, and (4) distributional assumptions for the likelihood and prior models. In addition, when additional sources of information about M_c become available, the proposed H-BMC method can be readily extended to incorporate such additional information in the forms of prior distributions.

Conclusions

Conventional methods for computing the completeness magnitude M_c rely on the FMD of earthquake catalogs, and they are known for several critical limitations in spatial estimation of M_c including gaps and large uncertainties of M_c in subregions of low seismicity. To overcome the limitations of the FMD-based methods for high-resolution M_c mapping, a decade ago, the BMC method was proposed to incorporate a priori information about M_c derived from its empirical relationship to the density of seismic stations, and since then it has been widely used.

In this article, we revisited the BMC method with a discussion on its critical shortcomings and proposed an HBM that inherently avoids these limitations for high-resolution M_c mapping, termed H-BMC. Using a published earthquake catalog from the Taiwan region, we demonstrated that, compared with the conventional FMD-based methods that are based only on observed magnitudes, the proposed H-BMC method can

effectively utilize a priori information about M_c in the form of a prior model and thereby fill gaps and reduce uncertainties in spatial mapping of M_c . We highlighted that the H-BMC method for spatial M_c mapping is free from certain restrictions and serves as a generic and flexible modeling framework for logically combining imprecise information about M_c from local observations and other sources.

Data and Resources

The earthquake catalog used in this study was originally recorded by the Taiwan Central Weather Bureau Seismic Network (CWBSN) and was relocated and made publicly available by Wu *et al.* (2008) at seismosoc.org/publications/BSSA_html/bssa_98-3/2007166-esupp (last accessed November 2021).

Declaration of Competing Interests

The authors acknowledge that there are no conflicts of interest recorded.

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